1. Bonding, Potential Wells & Properties

Equilibrium r_0: $F_N = F_A + F_R = 0$; E_N minimum ($-E_0$)

- **Melting T_m:** Deeper trough E_0 → higher bond energy → higher T_m.
- **Modulus E:** Curvature at trough: $E \propto (d^2E_N/dr^2)$ at r_0. Steeper = stiffer.
- **Expansion α_L:** Governed by well **asymmetry**. Symmetric well (deep) = low α_L (ceramics). Asymmetric = high α_L (polymers).
- **Bonds:** Ionic (electron transfer, nondirectional); Covalent (electron sharing, **directional**); Metallic (delocalized sea of e^-, ductile slip); Hydrogen (strongest secondary, ~50 kJ/mol).

2. Crystal Geometry & Theoretical Density

Cell	CN	n	a vs R	APF	Close Dir	Close Plane
FCC	12	4	a = 2R√2	0.74	⟨110⟩	{111}
BCC	8	2	a = 4R/√3	0.68	⟨111⟩	{110}
HCP	12	6	a=2R, c/a=1.633	0.74	⟨11-20⟩	(0001)

$\rho = (n \cdot A) / (V_c \cdot N_A)$ $[V_c = a^3, N_A = 6.022\times10^{23}]$

- Stacking: FCC = *ABCABC*... along [111]; HCP = *ABABAB*... along [0001].

3. Miller Indices, Directions & Planes

- **Directions [uvw]:** Tail at origin; head in a,b,c; clear fractions; overbar for negative. Family: ⟨uvw⟩.
- **Planes (hkl):** **NEVER pass through origin!** (Shift origin if needed). Read intercepts A,B,C; take reciprocals (1/A, 1/B, 1/C; 1/∞=0); clear fractions. Family: {hkl}.
- **Hexagonal (hkil):** 4 axes (a1, a2, a3 at 120°, z). Always: **i = -(h + k)**.
- **Densities:** LD = atoms on line / length L; PD = 2D atoms in plane / area A.

4. XRD & Bragg's Law Selection Rules

Bragg's Law: $n\lambda = 2\, d_{hkl} \sin\theta$ | Cubic: $d_{hkl} = a / \sqrt{(h^2 + k^2 + l^2)}$

- **BCC Peak Rule:** h + k + l must be **EVEN**: (110), (200), (211), (220). (100) and (111) are forbidden!
- **FCC Peak Rule:** h, k, l must be **ALL ODD or ALL EVEN**: (111), (200), (220), (311). (110) and (211) are forbidden!

5. Defects & Hume-Rothery Rules

Vacancies: $N_v = N \exp(-Q_v / kT)$ $[k = 8.62\times10^{-5}$ eV/K]

- **Hume-Rothery (Complete Solid Solubility, Cu-Ni):** 1. Size: $|\Delta r|/r_{solvent} < 15\%$ · 2. Crystal: identical (both FCC) 3. Electronegativity: comparable · 4. Valency: higher dissolves more.

6. Slip Systems & Schmid's Law

- **Edge:** Extra half-plane; Burgers b ⊥ line. **Screw:** Spiral ramp; b ∥ line.
- **FCC Slip:** {111}⟨110⟩ → 4×3 = **12 systems** (high ductility).
- **BCC Slip:** {110}⟨111⟩ → 6×2 = 12 systems (total 48; high strength).

Schmid: $\tau_R = \sigma \cos\varphi \cos\lambda$ | Max slip at $\varphi=\lambda=45°$: $\sigma_y = 2\,\tau_{CRSS}$

- **Strengthening:** 1. Hall-Petch: $\sigma_y = \sigma_0 + k_y d^{-1/2}$ (finer grains = stronger) · 2. Solid solution pinning · 3. Cold work: $\%CW = (A_0 - A_d)/A_0 \times 100\%$ (increases σ_y and UTS, reduces %EL).

7. Diffusion (Fick's Laws)

- **Mechanisms:** Interstitial (C in Fe) is *much faster* than Vacancy. Cu-Ni couple undergoes **interdiffusion** (both diffuse simultaneously).
- **Fick 1st (Steady):** $J = -D\,(dC/dx) = -D\,(C_A - C_B)/(x_A - x_B)$.
- **Arrhenius:** $D = D_0 \exp(-Q_d / RT)$ → ln D vs 1/T slope = $-Q_d/R$.
- **Fick 2nd (Semi-Infinite):** $(C(x,t) - C_0)/(C_s - C_0) = 1 - \text{erf}(x / 2\sqrt{(Dt)})$.
- **Scaling:** Constant composition → $x^2 / (D \cdot t)$ = constant.

8. Mechanical Properties & Stress-Strain

$\sigma = F/A_0$ [MPa] | $\varepsilon = \Delta l/l_0$ | $\sigma = E\varepsilon$ (Hooke) | $\nu = -\varepsilon_x/\varepsilon_z \approx 0.33$

- **Elastic Modulus E:** Depends on bond stiffness ONLY. Cold work, grain size, and alloying do NOT change E! (Steels E ≈ 207 GPa).
- **Point B:** Proportional limit. **Point C:** 0.2% offset yield (σ_y).
- **Point E (UTS):** Max stress. **Necking begins strictly AFTER Point E!**
- **Uniform plastic:** Point C to E. **Non-uniform (necking):** Point E to fracture.
- **True Stress/Strain (up to necking):** $\sigma_T = \sigma(1 + \varepsilon)$, $\varepsilon_T = \ln(1 + \varepsilon)$.
- **Resilience:** $U_r \approx \sigma_y^2 / (2E)$. **Toughness:** Total area under σ-ε curve.